

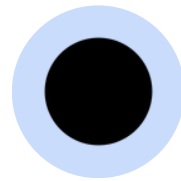
DATA SCIENCE CASE STUDY

Loan Offer Propensity Model

Chuene Mosomane

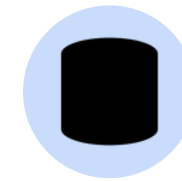


What we'll cover



Business Problem

The bank wants to offer their customers a product - loan. It wants to identify which customers to approach, and if there is any bias in the data.



Data & EDA findings

We have three tables:

- customers
- loans
- transactions

What the three source actually looks like.



Propensity Modelling

- I develop a 2 levels-framework - customer and loan levels
- Then build a classification model to predict the probability of a which customers will take up a loan.
- The customers identified are stored in a CSV file



Feature engineering

I engineered financial features to check the financial-health KPIs behind the model, in plain terms:

- Net cash flow, balance volatility, low balance frequency
- Balance to income ratio, burn rate.



Model & fairness results

- I performed feature selection to rank discriminant features.
- Then train and evaluate a Logistic regression with 85/15 split.
- Lastly generate a fairness report: approval by age and gender



MLOps: pipeline, monitoring, next steps

- The solution is written in PySpark and runs in production.
- Monitoring for key metrics is done in Grafana dashboards
- Recommendations: This solution is used by the marketing team for promotional intervention.

Which customers should we approach with a loan offer?

Objective

Use exploratory analytics, visualization, and machine learning on customer, transaction, and loan data to identify clients we should proactively approach with a loan offer.

Why it matters

- Targeted offers convert better than blanket campaigns and cost less to run
- Getting this wrong at scale means either missed revenue or reputational/regulatory risk
- A model that can't explain *why* it flags someone, it is a liability in a regulated lending context

Three tables, one customer ID to join them

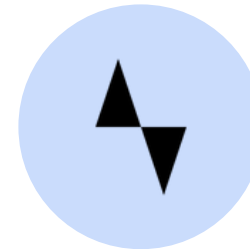
```
customers: 2500 rows  
loans: 3203 rows  
transactions: 91931 rows
```



Customers

- Gender
- Age
- Income

One row per client



Transactions

- Date
- Amount (+in / -out)
- Balance after transaction

Many rows per client, over time



Loans

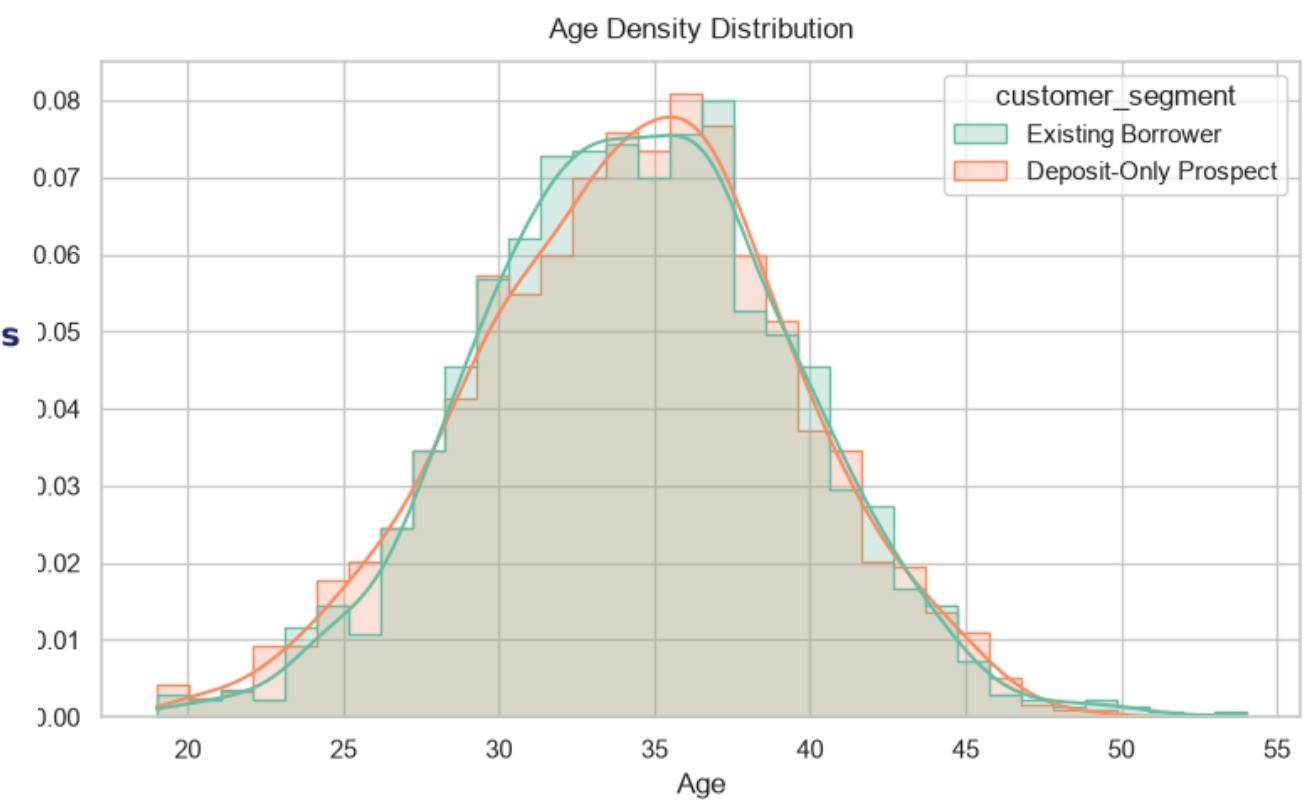
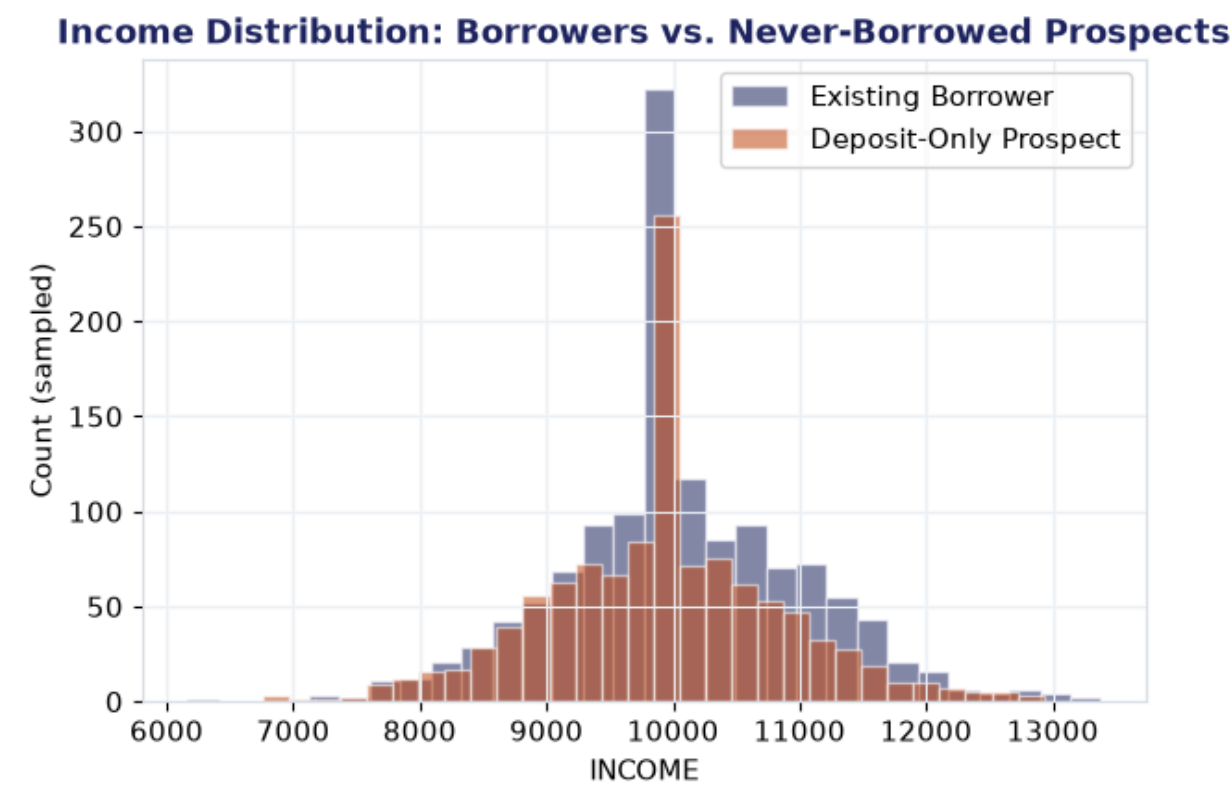
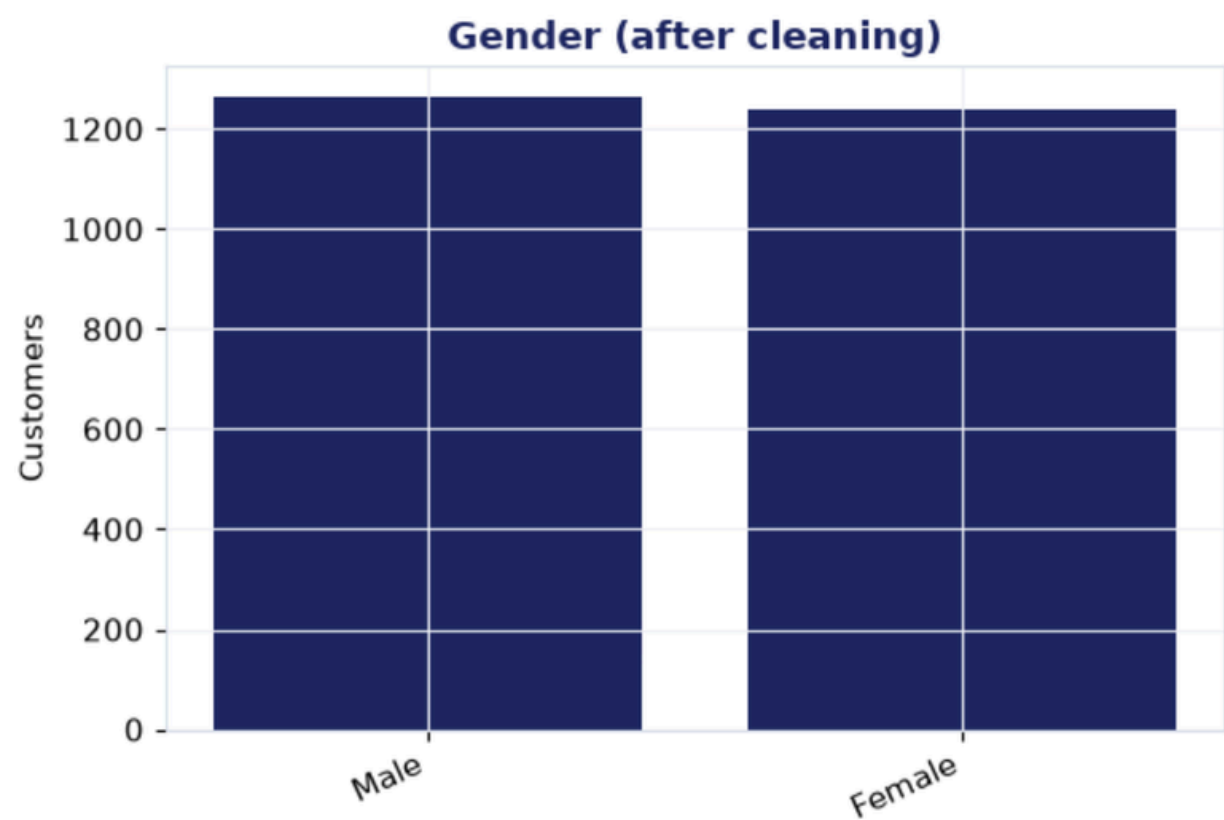
- Date applied
- Amount requested
- Outcome: TakeUp / Declined

One row per historical application

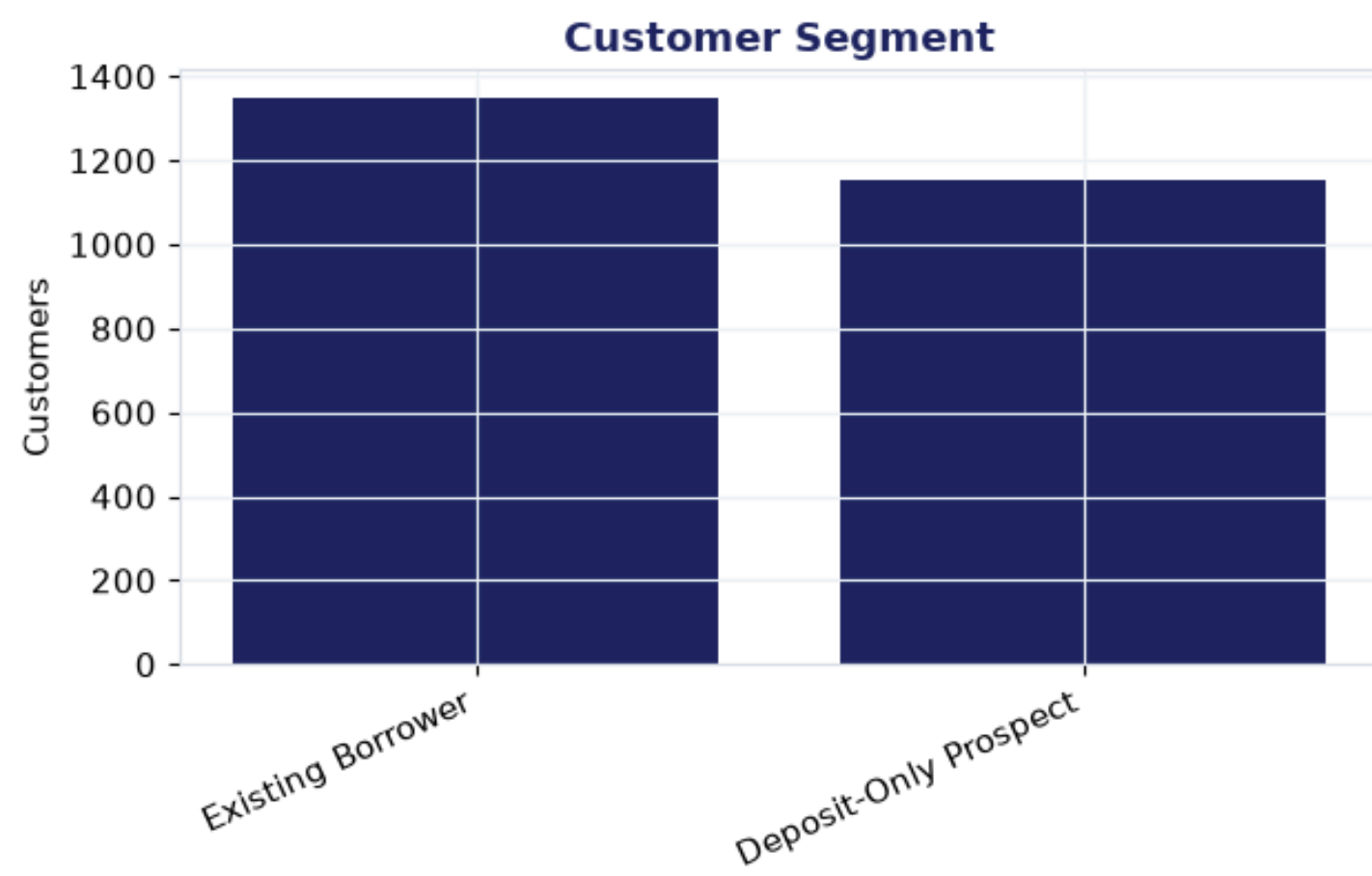


Joined on customer ID into two working tables: a customer-level view (for EDA and segmentation) and a loan-application-level view (for modelling — see slide 6).

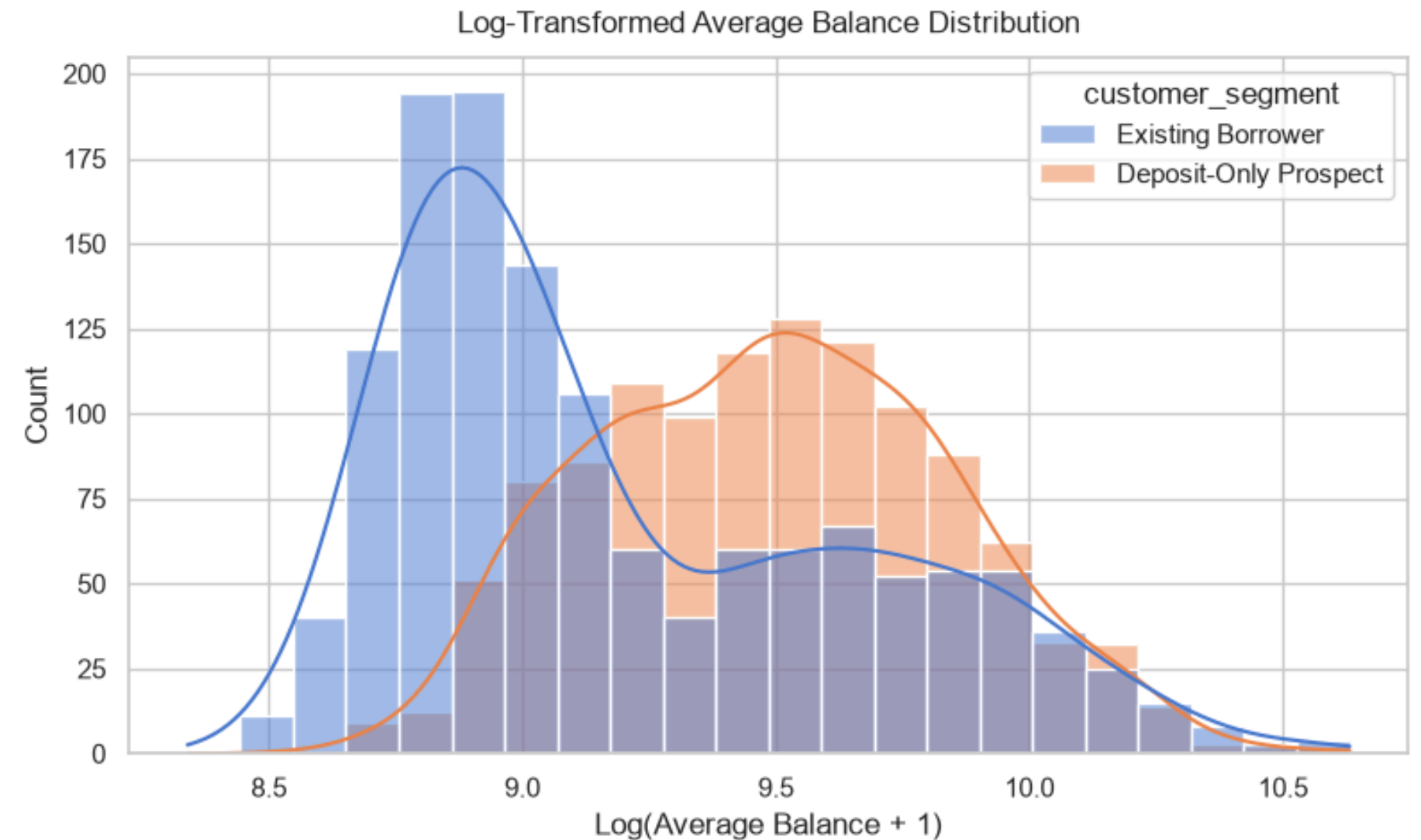
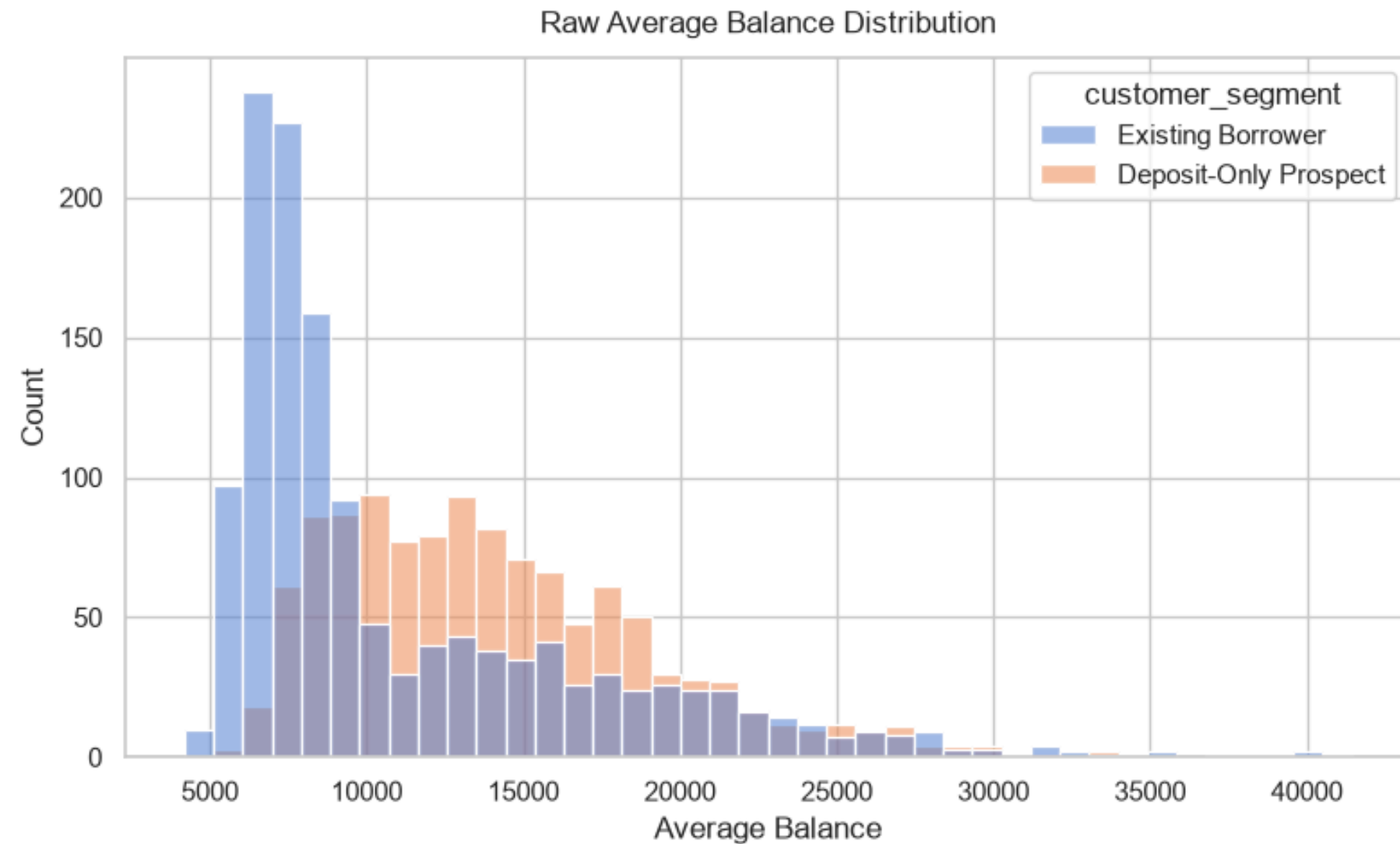
What are the Demographics?



- Our customers consists of a fair distribution of male and female who earns ~R10000 and mostly are youth as age peaks around 35 years.

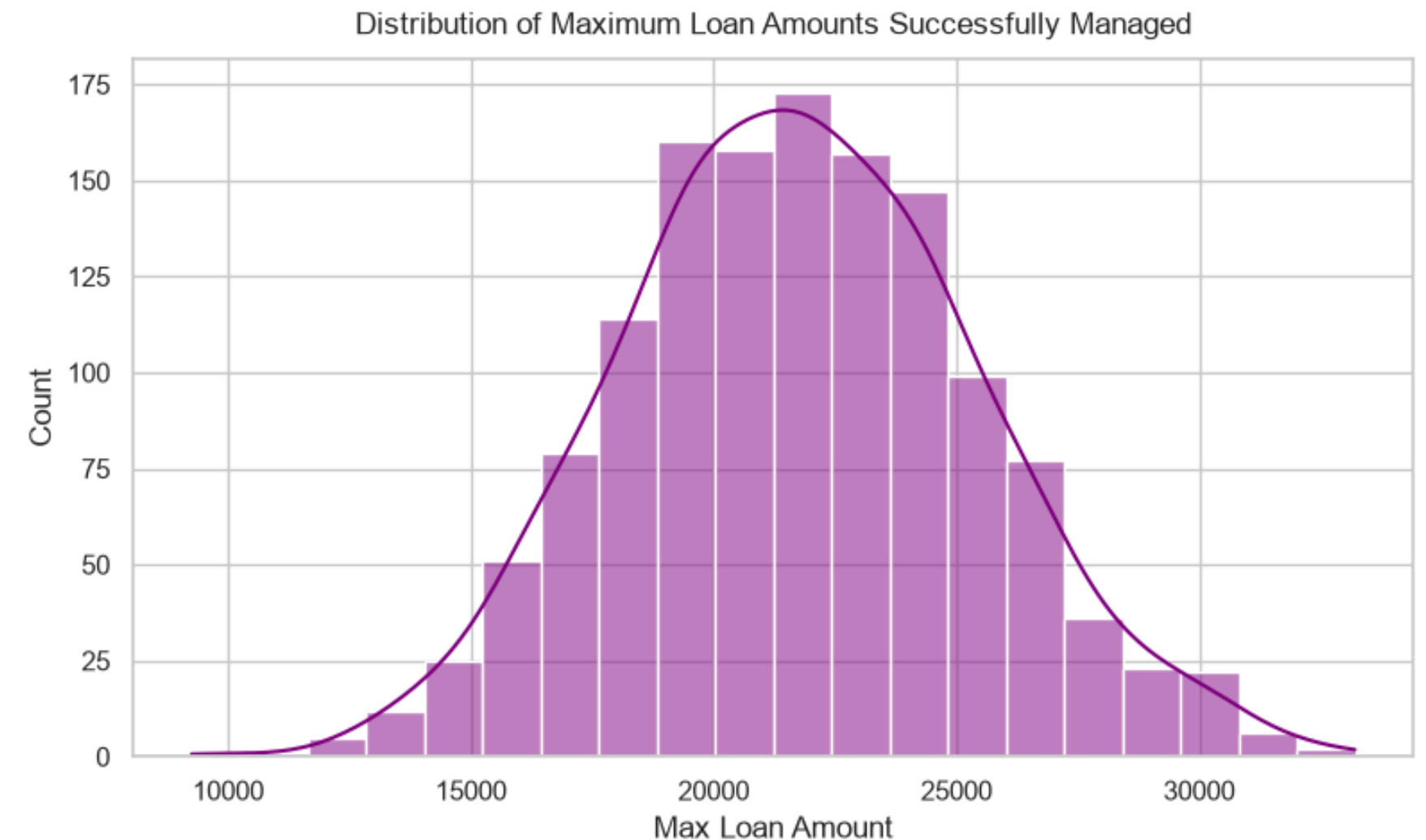
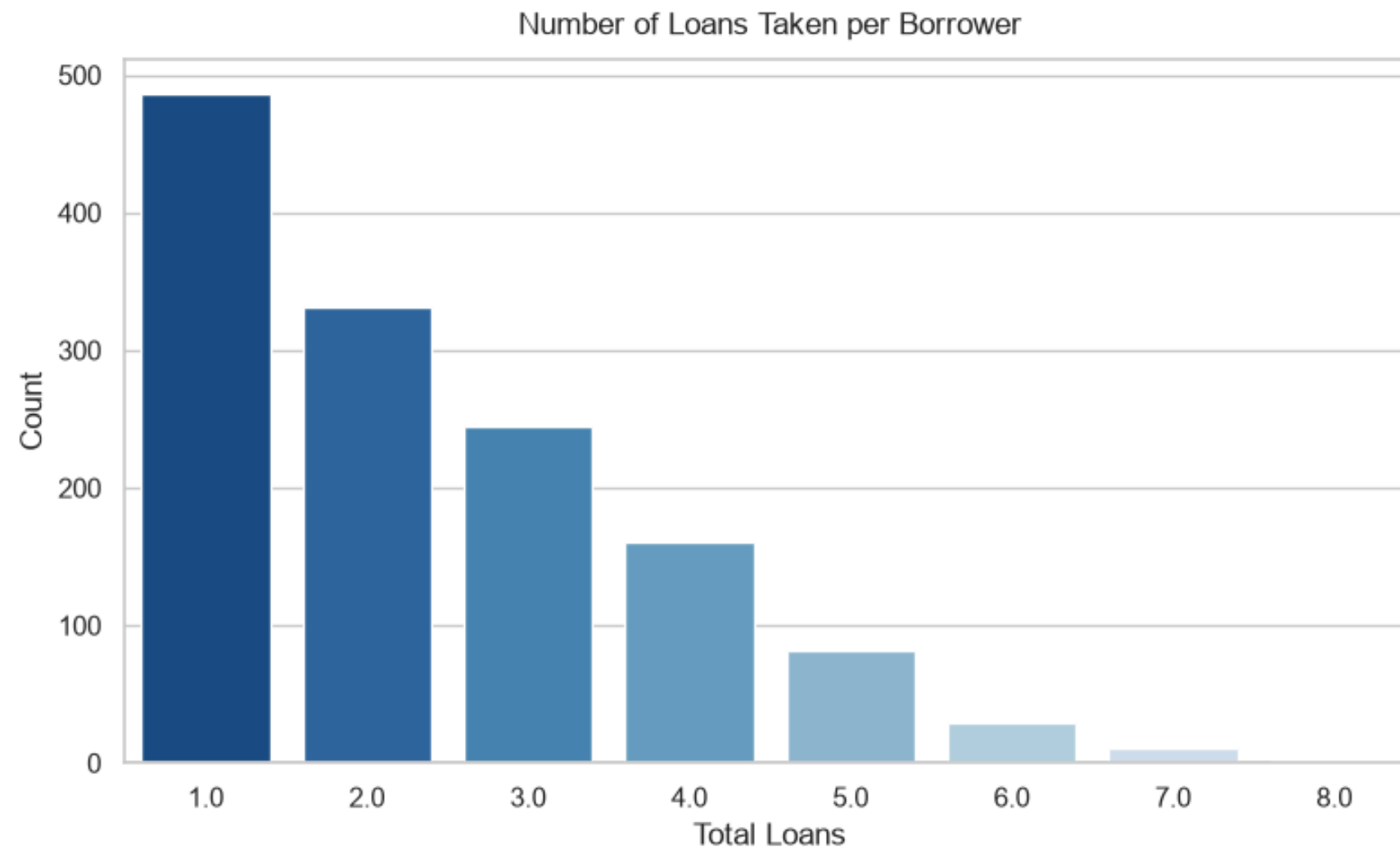


What is the disposable cash our customers have?



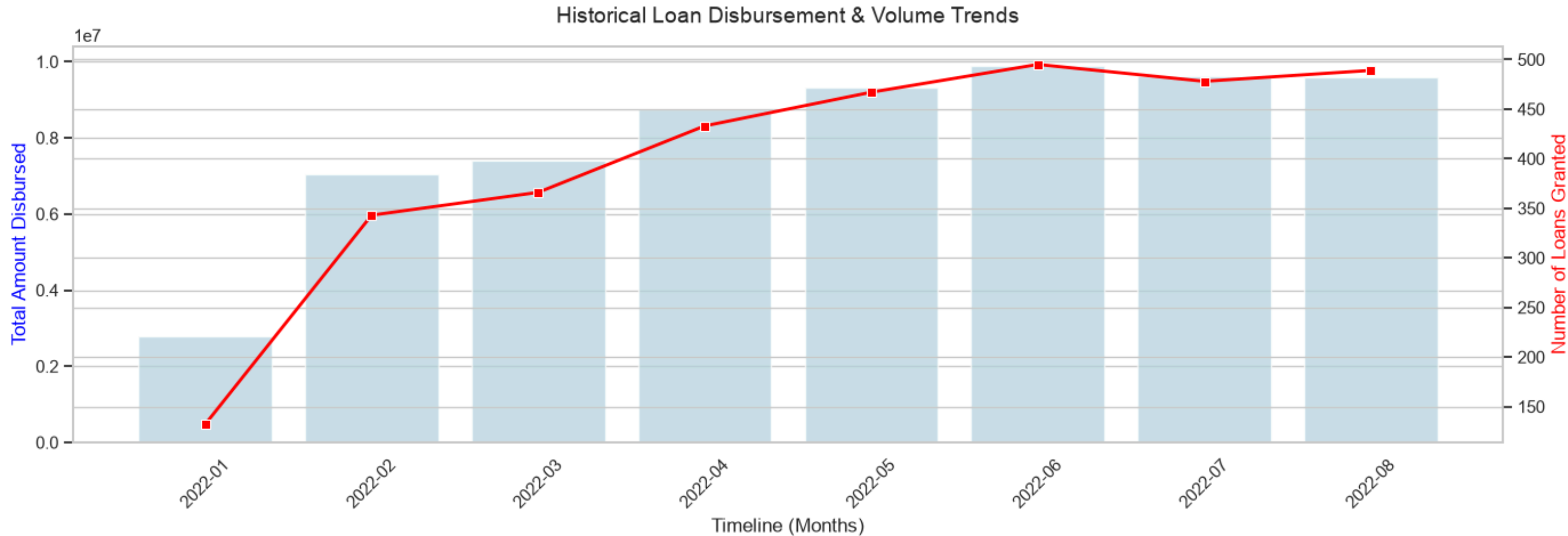
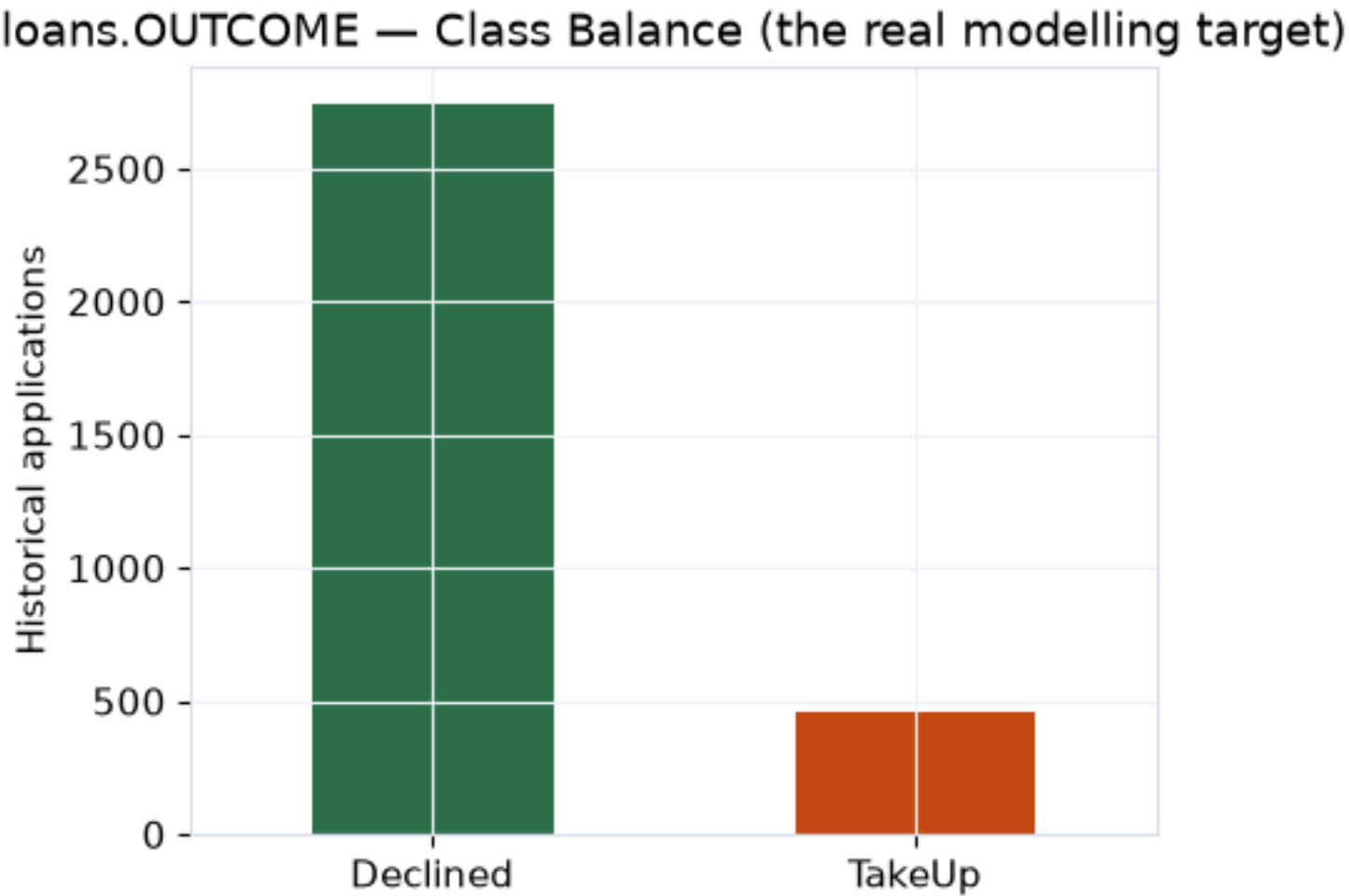
- We see that most of existing borrower customers have disposable cash of ~R8000 and only around 10% of our customers have higher disposable cash ~R25000
- When we zoom in, we further see that the deposit-only prospect customers have the same disposable cash as the existing borrowers customers at the tail.

Does our customers take loans and how much?



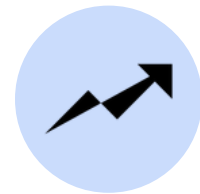
- Most customers take up a single loan of around R21000.
- I shall use this feature because it has a normal distribution shape, to engineer KPI features

How many customers take-up a loan and what impact does it have on business?



Even though it looks like there is less number of people taking up the loan but there is a increase on the number of loans that are granted.

KPIs behind the model



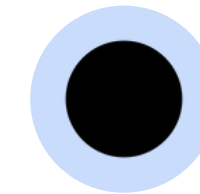
Net cash flow

Average monthly inflow minus outflow — a steady positive gap signals capacity to repay.



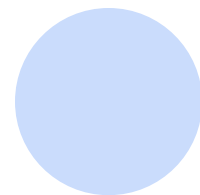
Balance volatility

Std. dev. of balance vs. its average — a shaky balance suggests higher near-term risk.



Burn rate

How fast balance drops after it peaks each month (a proxy for “just got paid”) — currency lost per day.



Low-balance frequency

Count of near-zero balance events — threshold set from the data's own 10th percentile, not a guessed round number.



Credit capacity

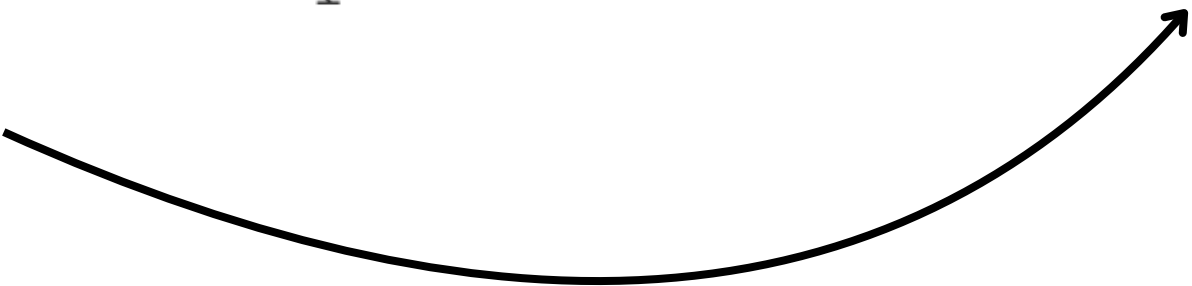
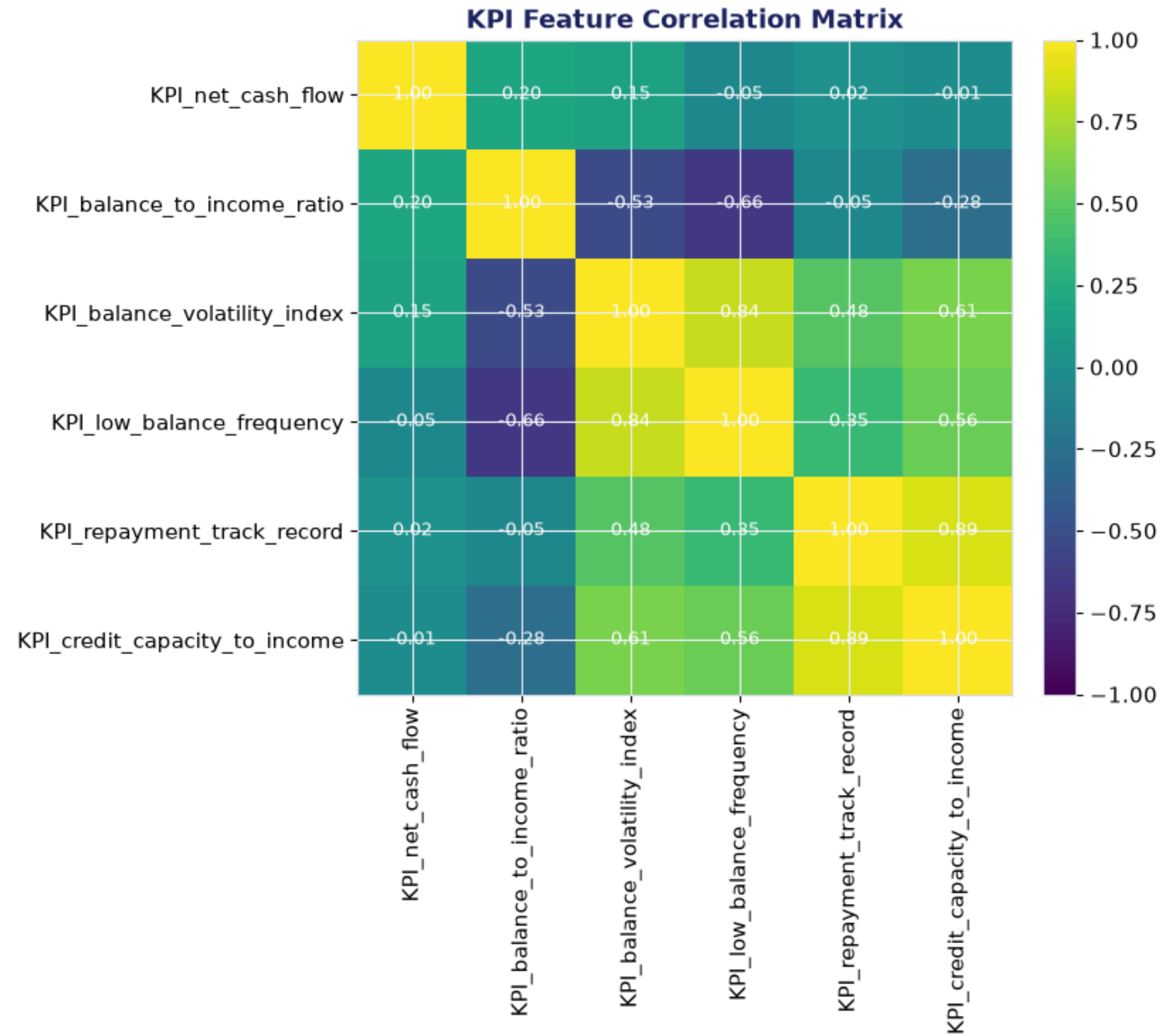
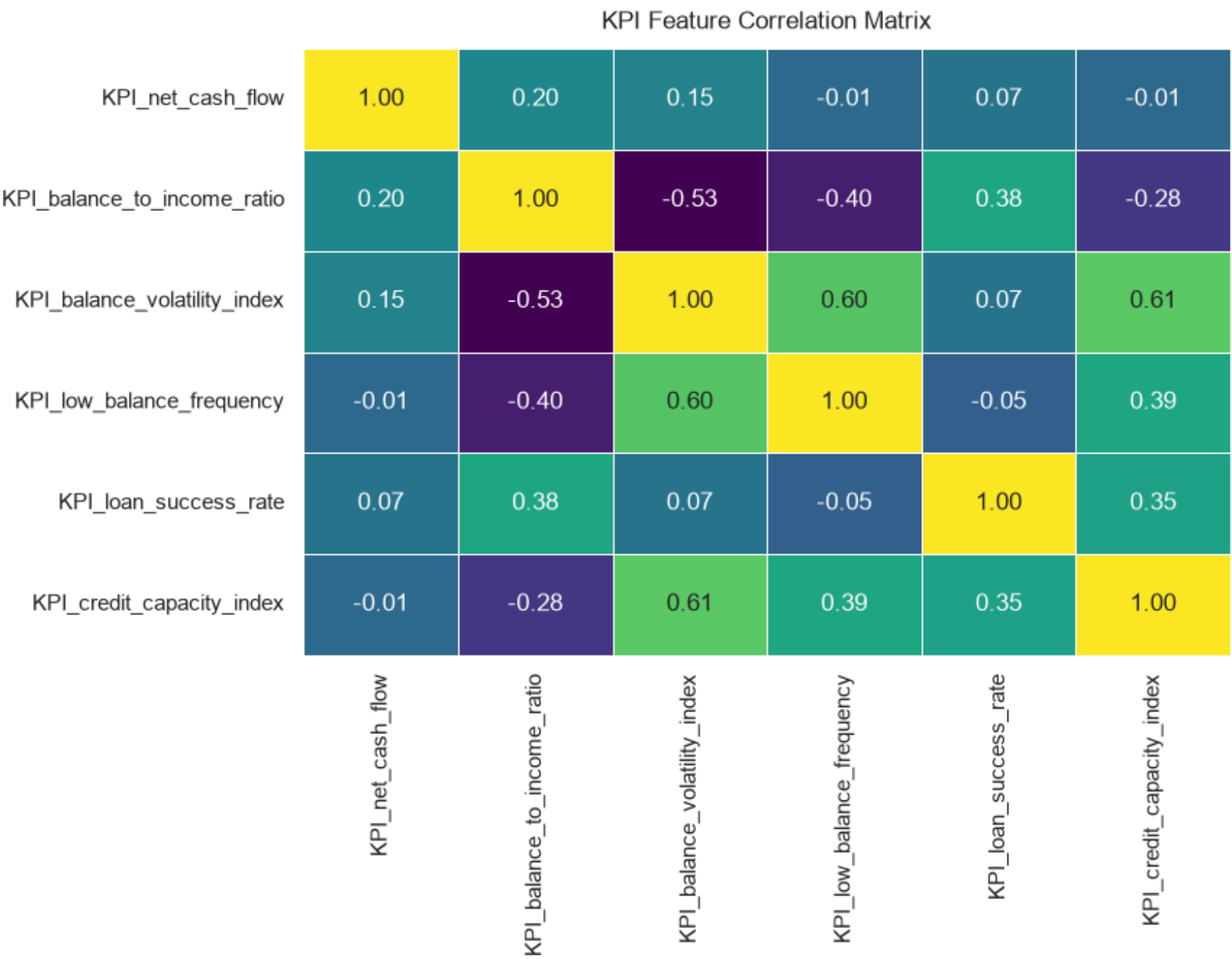
Largest amount ever successfully taken up, relative to income — a proven, not assumed, borrowing ceiling.



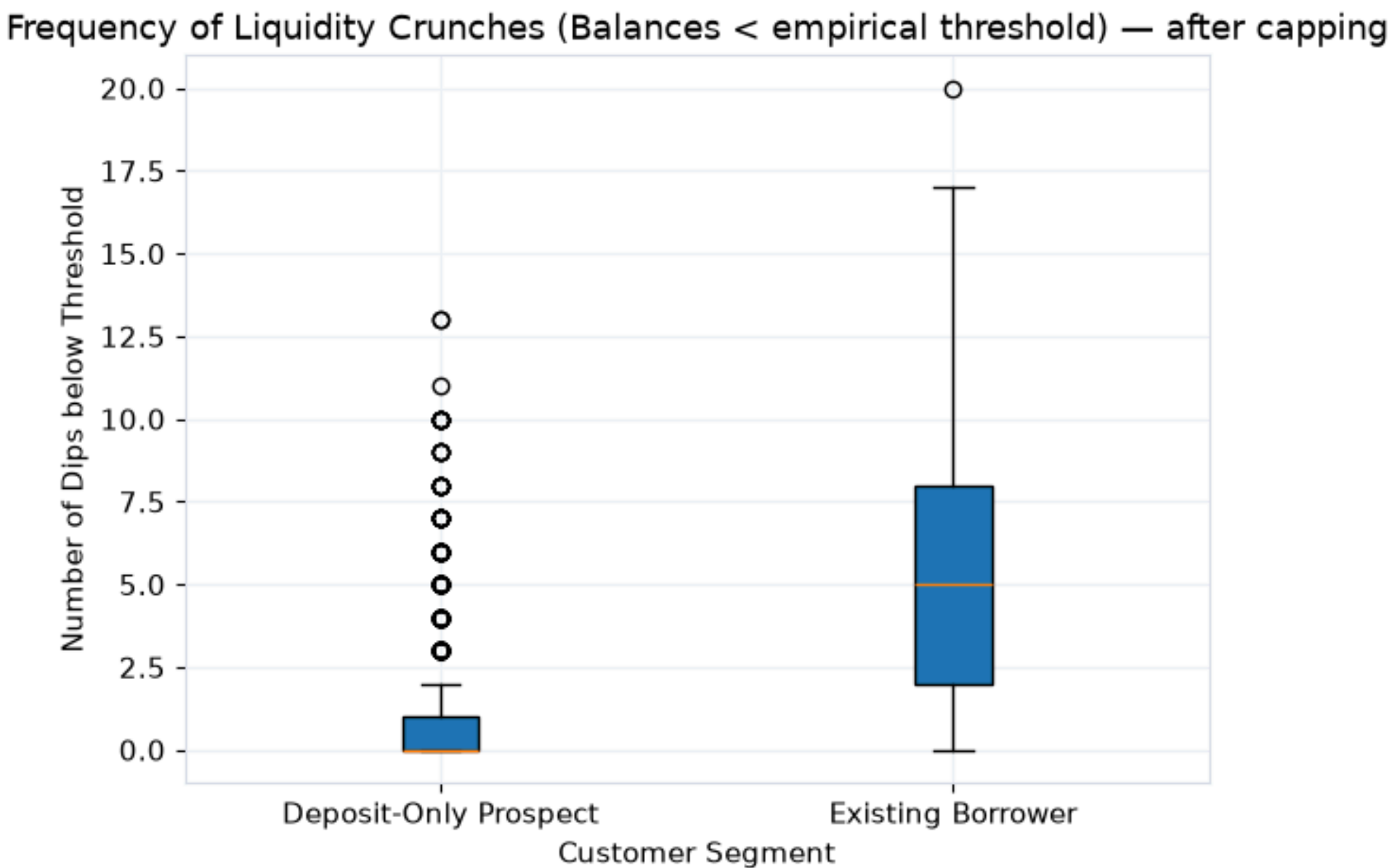
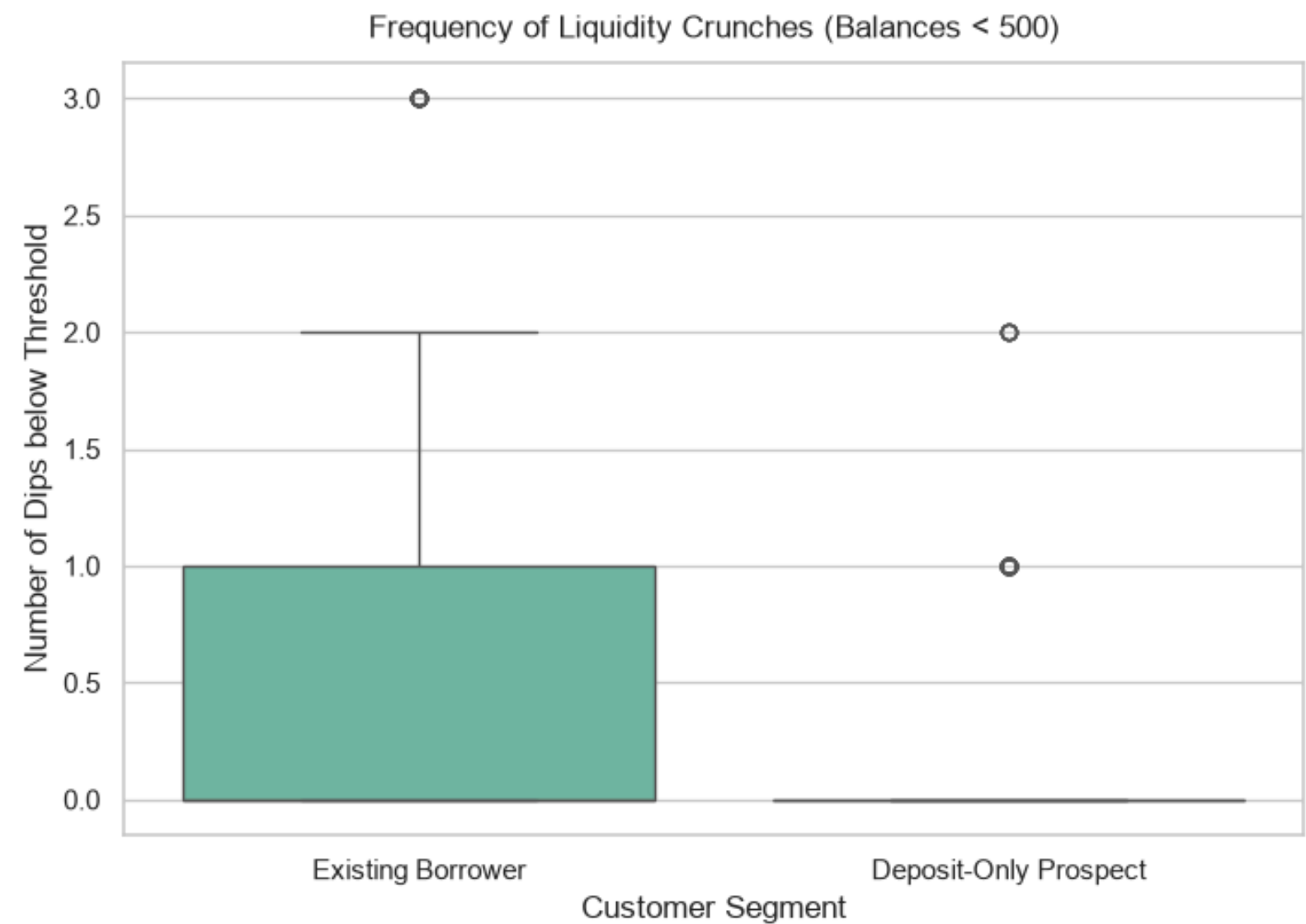
Lifecycle segment

Age × income quadrant (tertile cut-offs from the data) — young/low-income vs. mid-age/high-income vs. senior, etc.

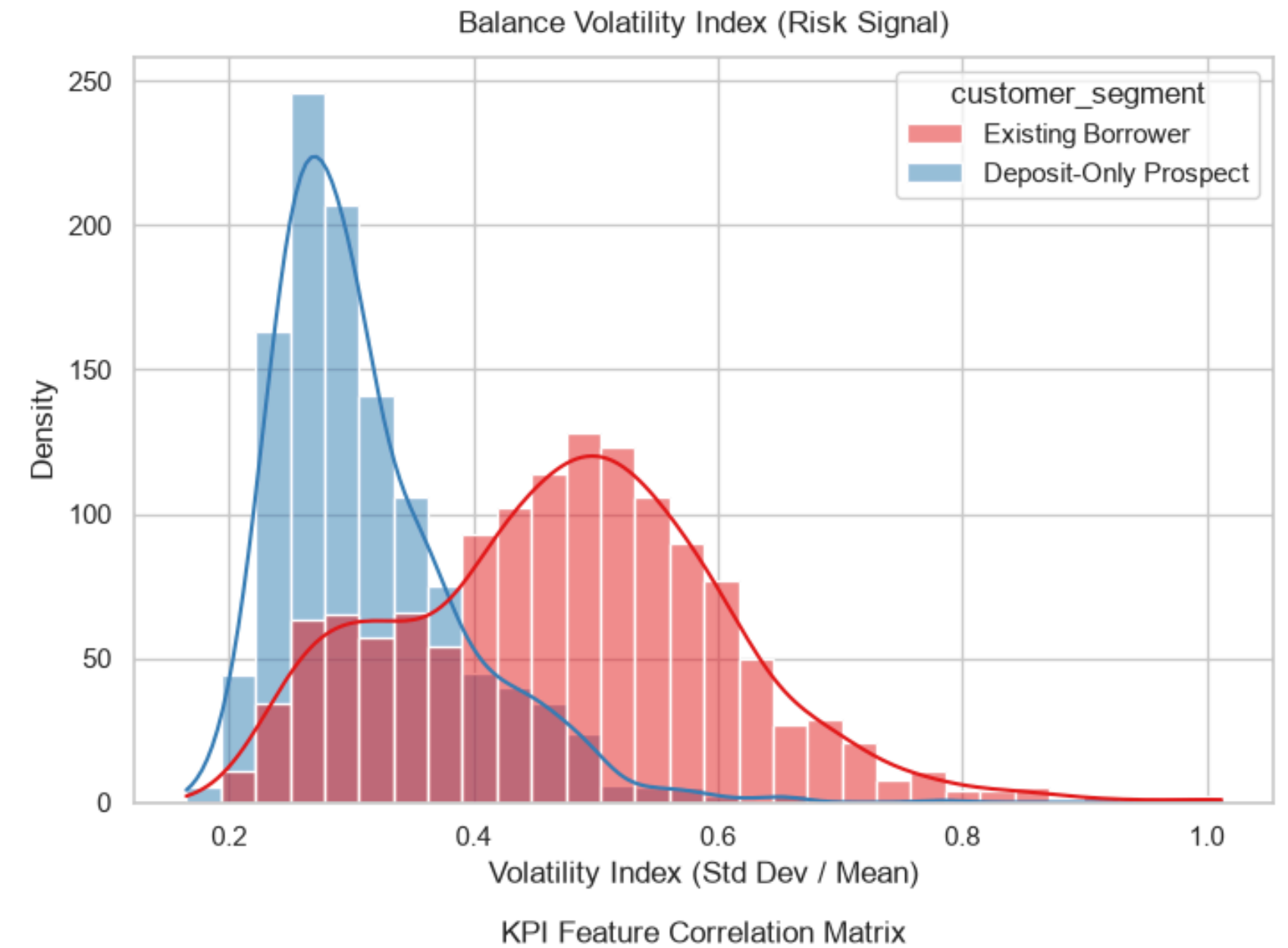
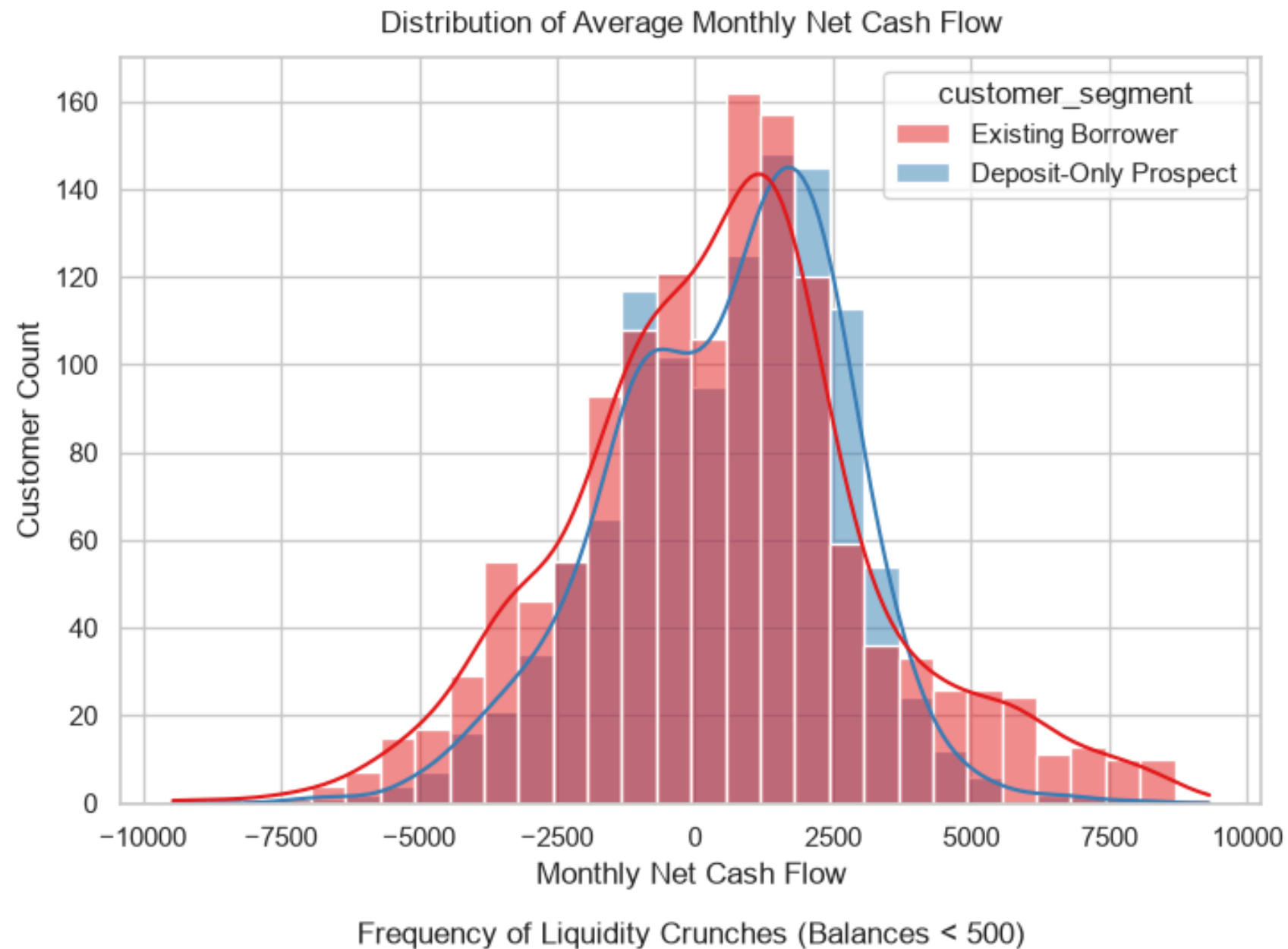
Dealing with Colinearity !



How often does the balance go to negative status?

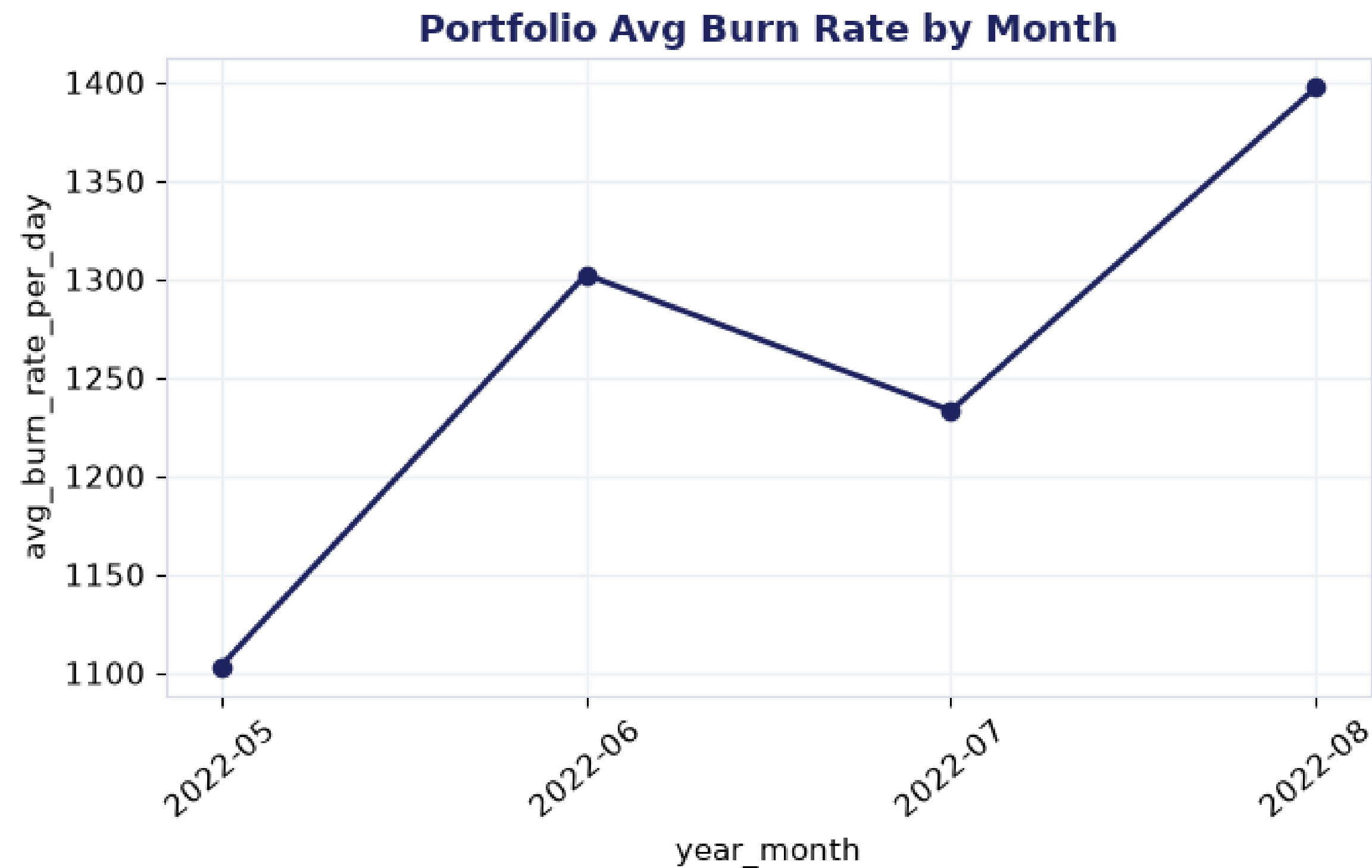


What is cash flow and balance volatility?

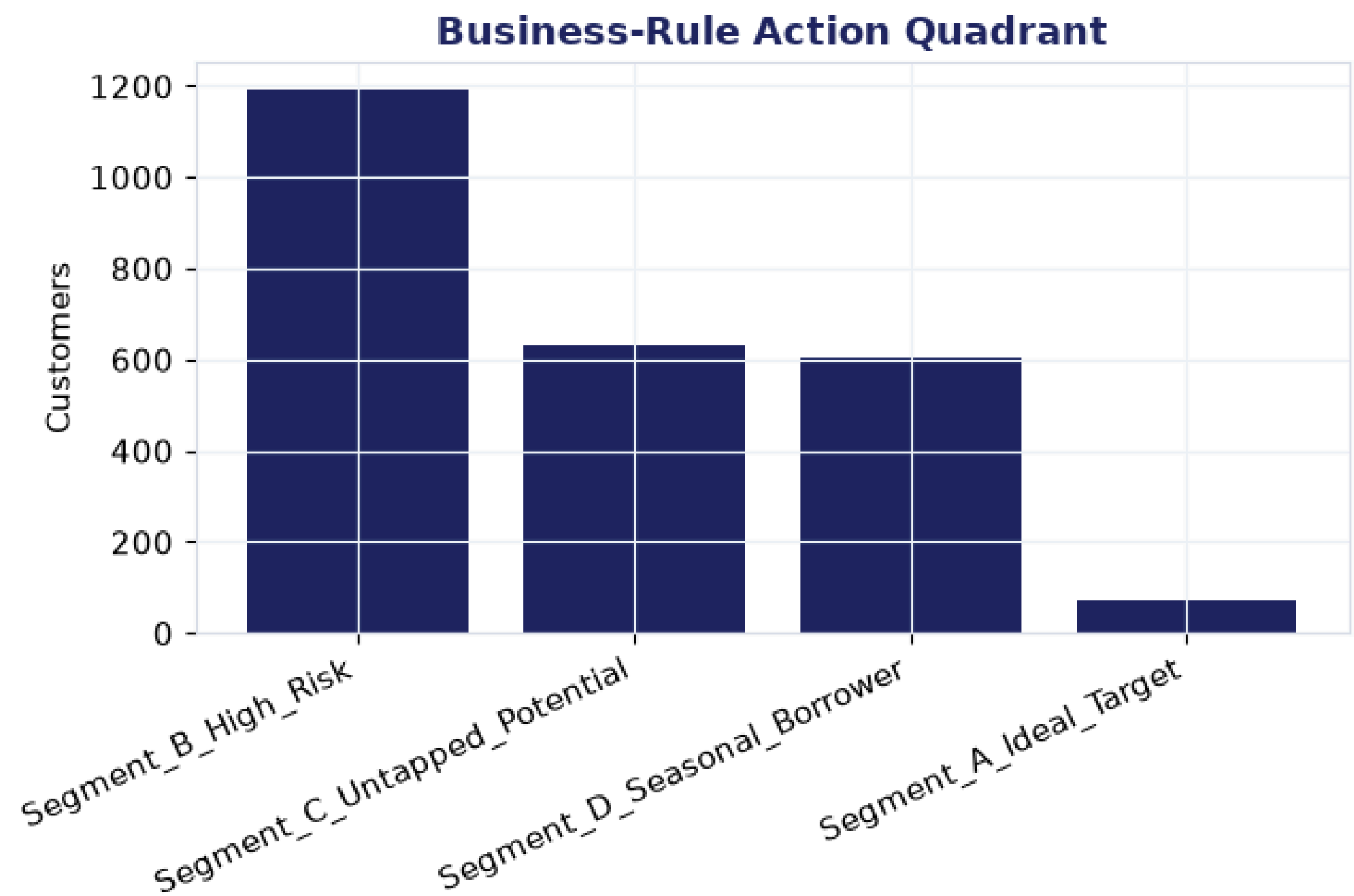


- Even though our customers have a positive cash flow, the majority of our customers - existing borrower, their balance is more volatile.

What is the burn rate of our customers

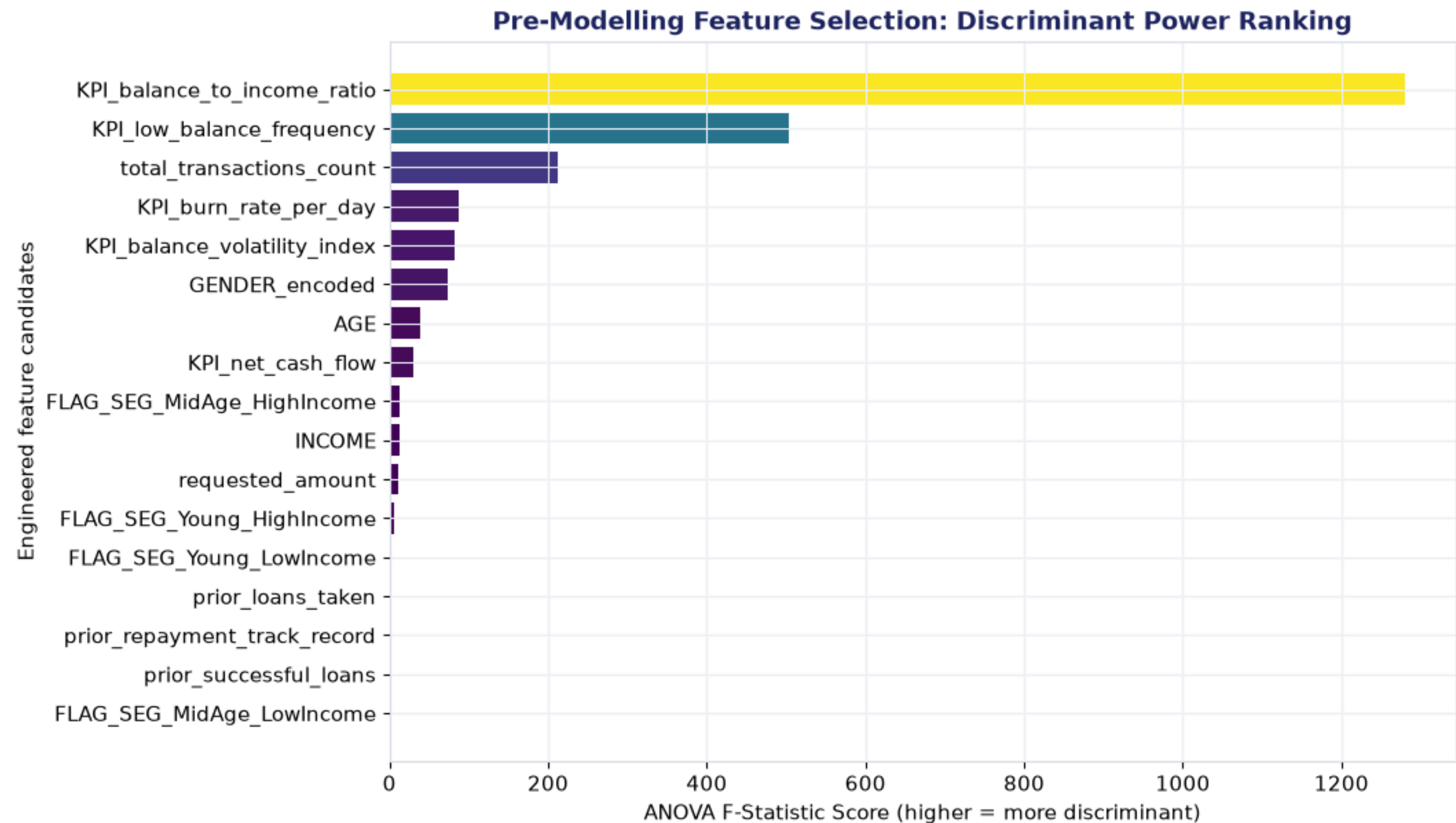


Customer Segmentation



- Now we know enough about our customers and we proceed to segment them into risk using KPI's

Feature Selection



Model & evaluation approach

Methodology

Algorithm:

Logistic Regression — interpretable coefficients matter more and therefore we choose it as a baseline

Class imbalance:

Inverse-frequency row weighting (Spark ML's equivalent of scikit-learn's `class_weight='balanced'`)

Split:

Stratified 80/20 train/test, split independently per class to preserve the outcome ratio in both halves

Grain:

One row per historical loan application — not per customer

Headline results

ROC-AUC (test set)

90%

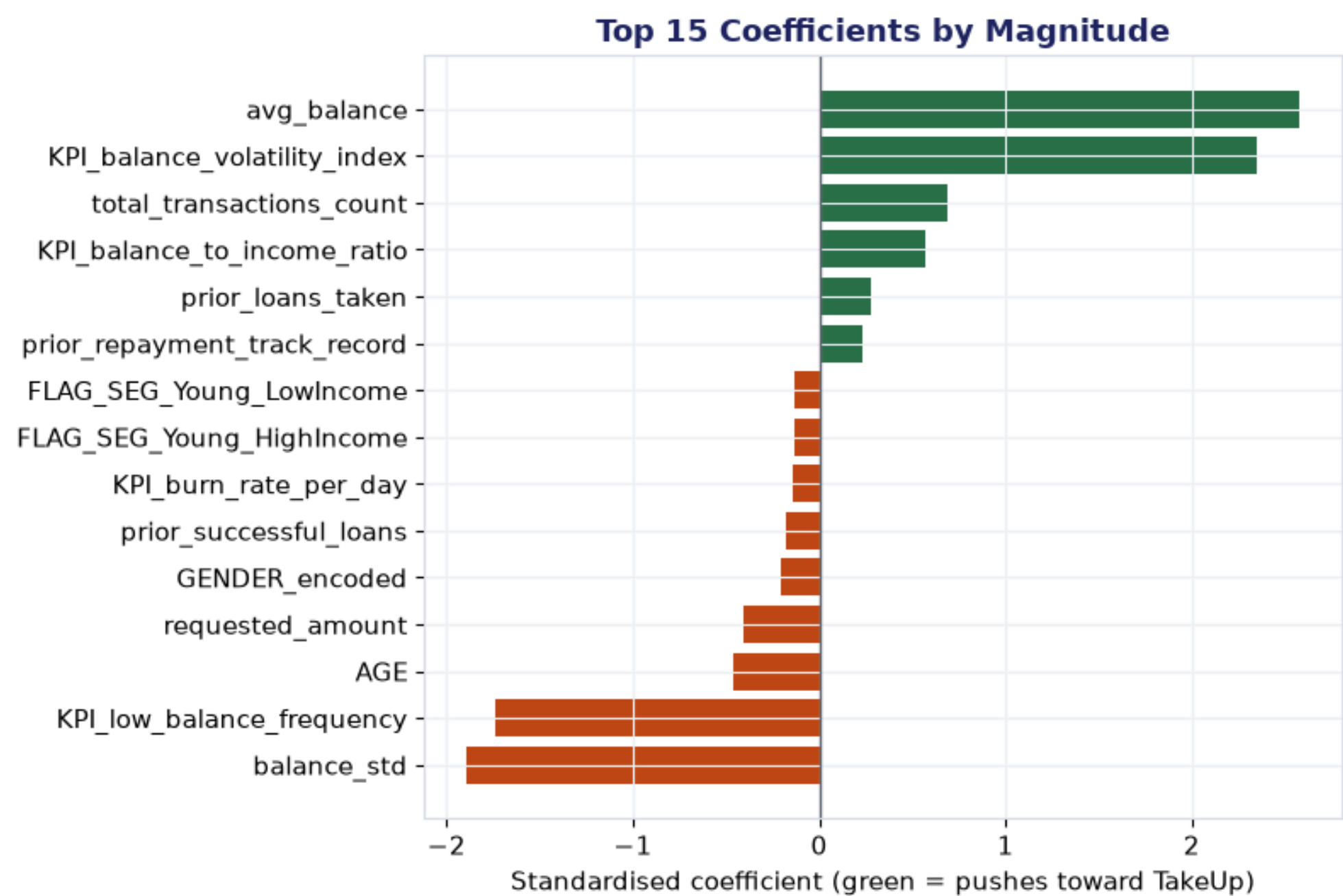
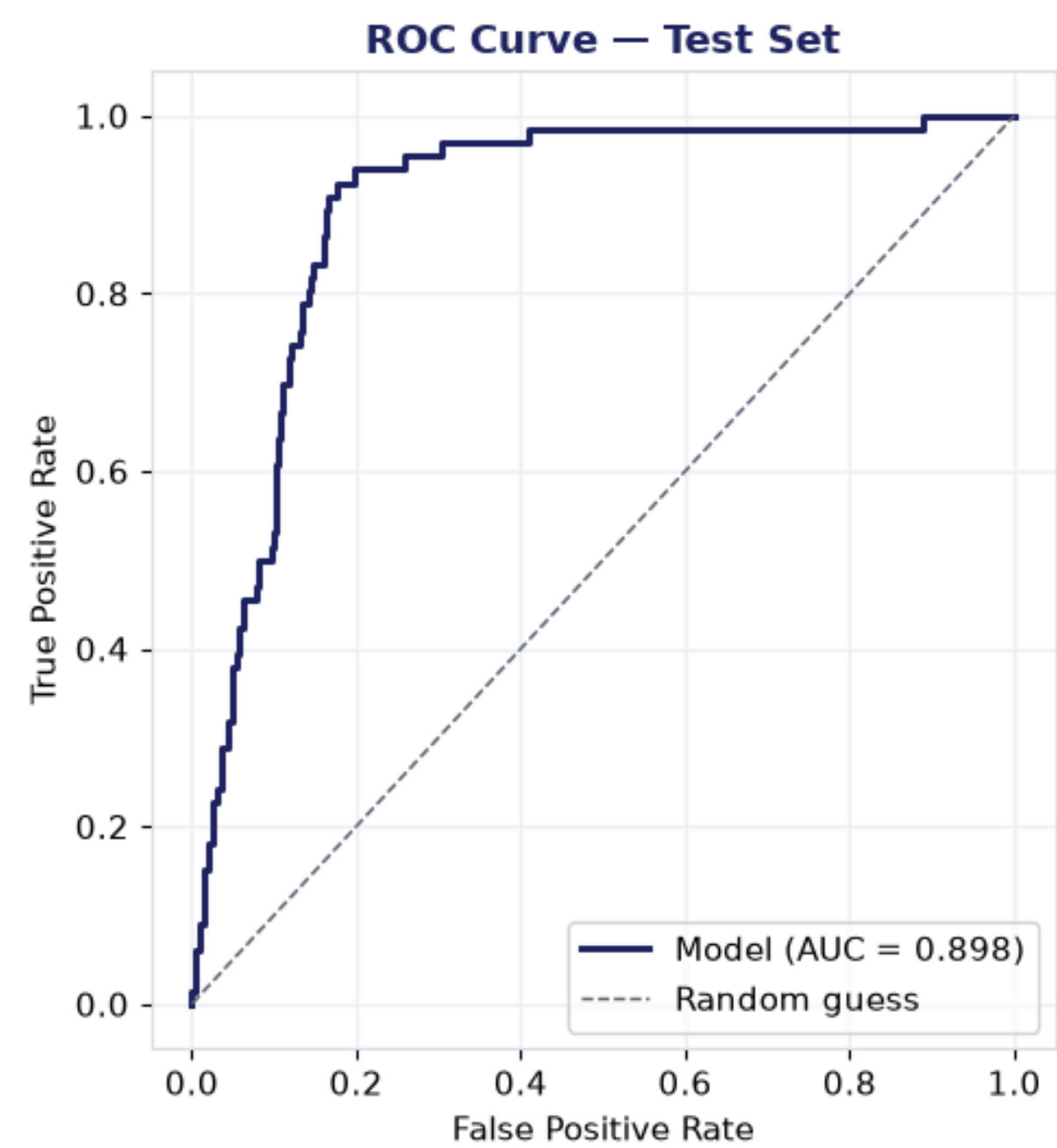
AUC without GENDER

86%

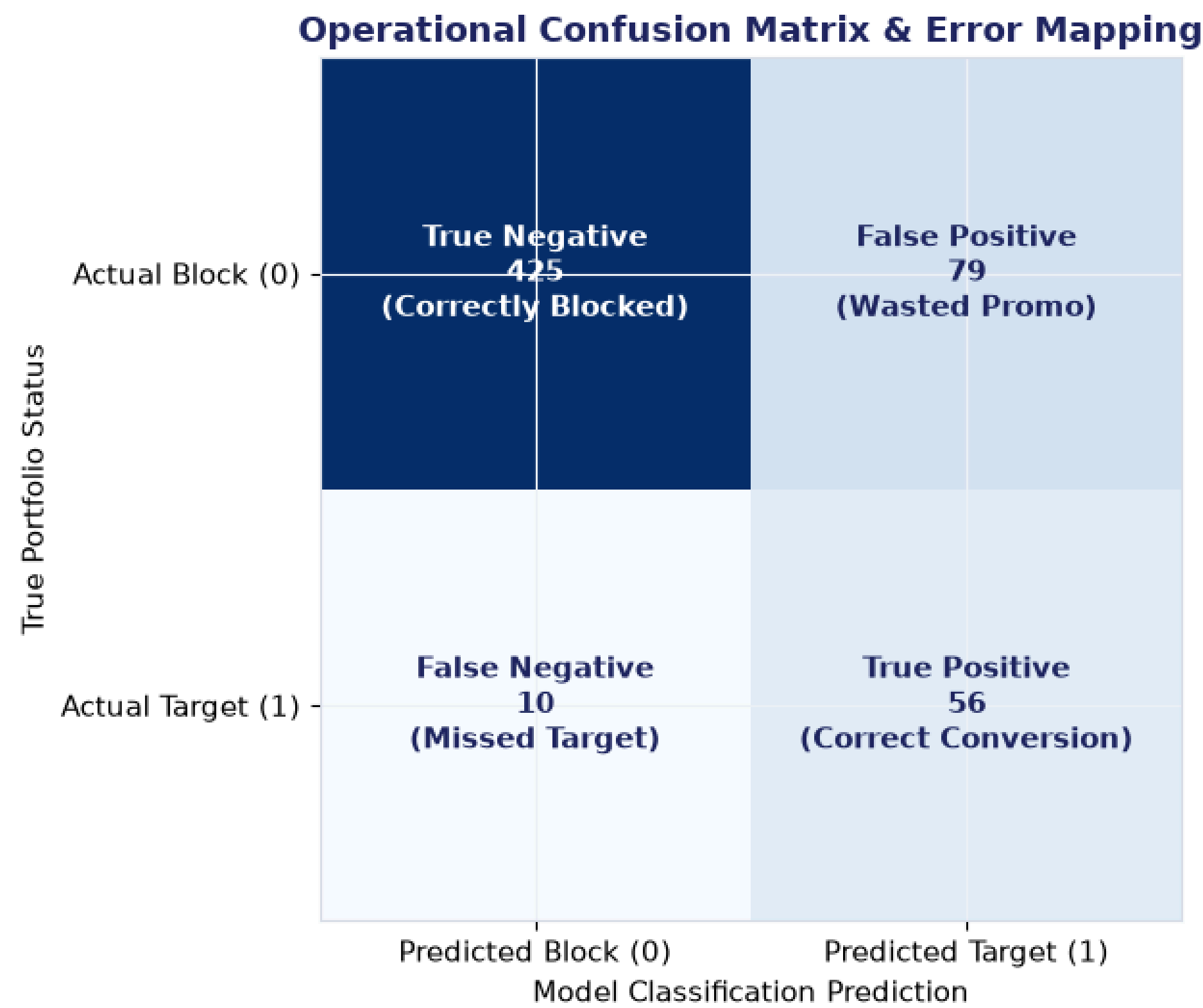
Train / test size

85/15

Model Results



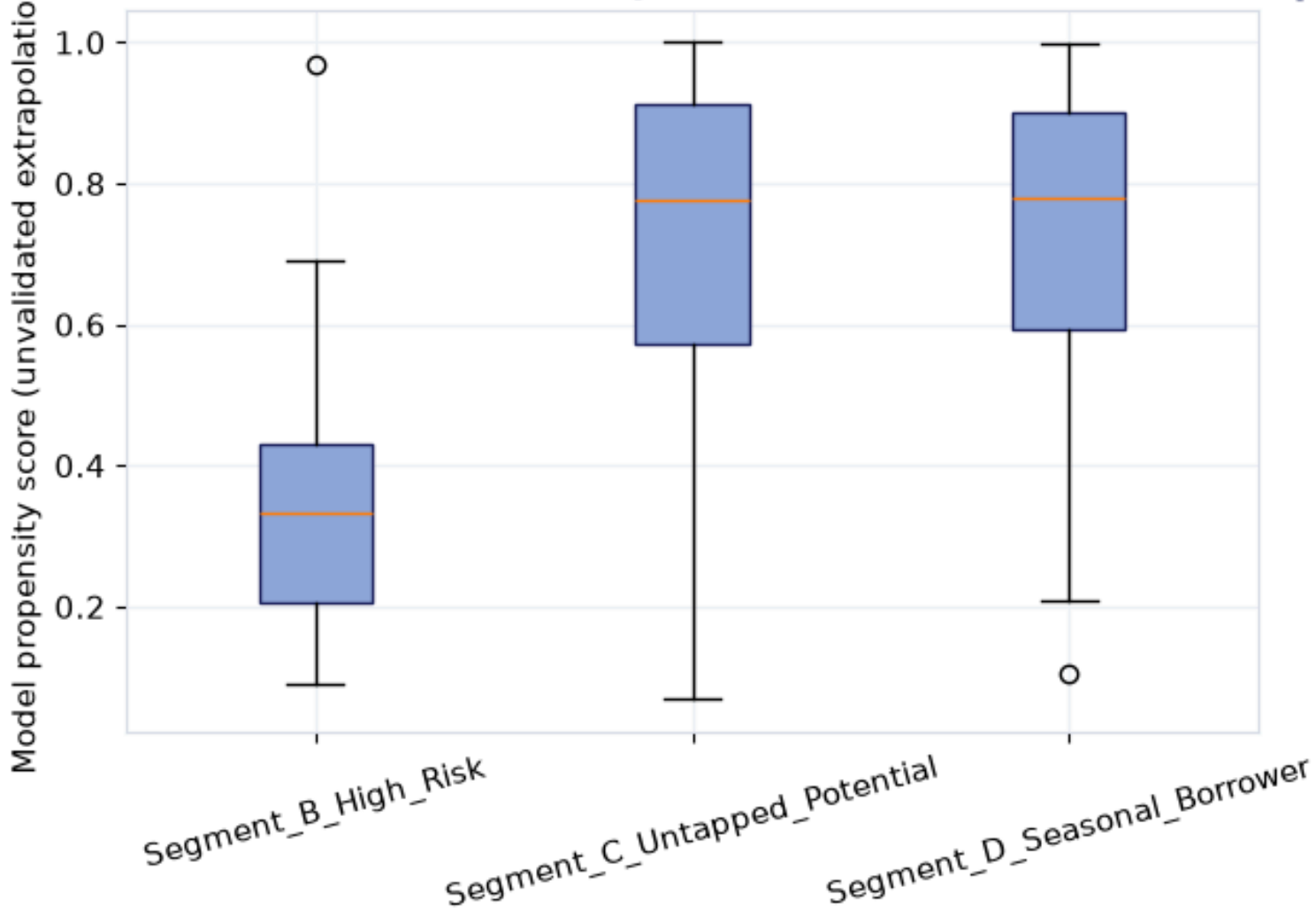
How often are we correct?



Identified List

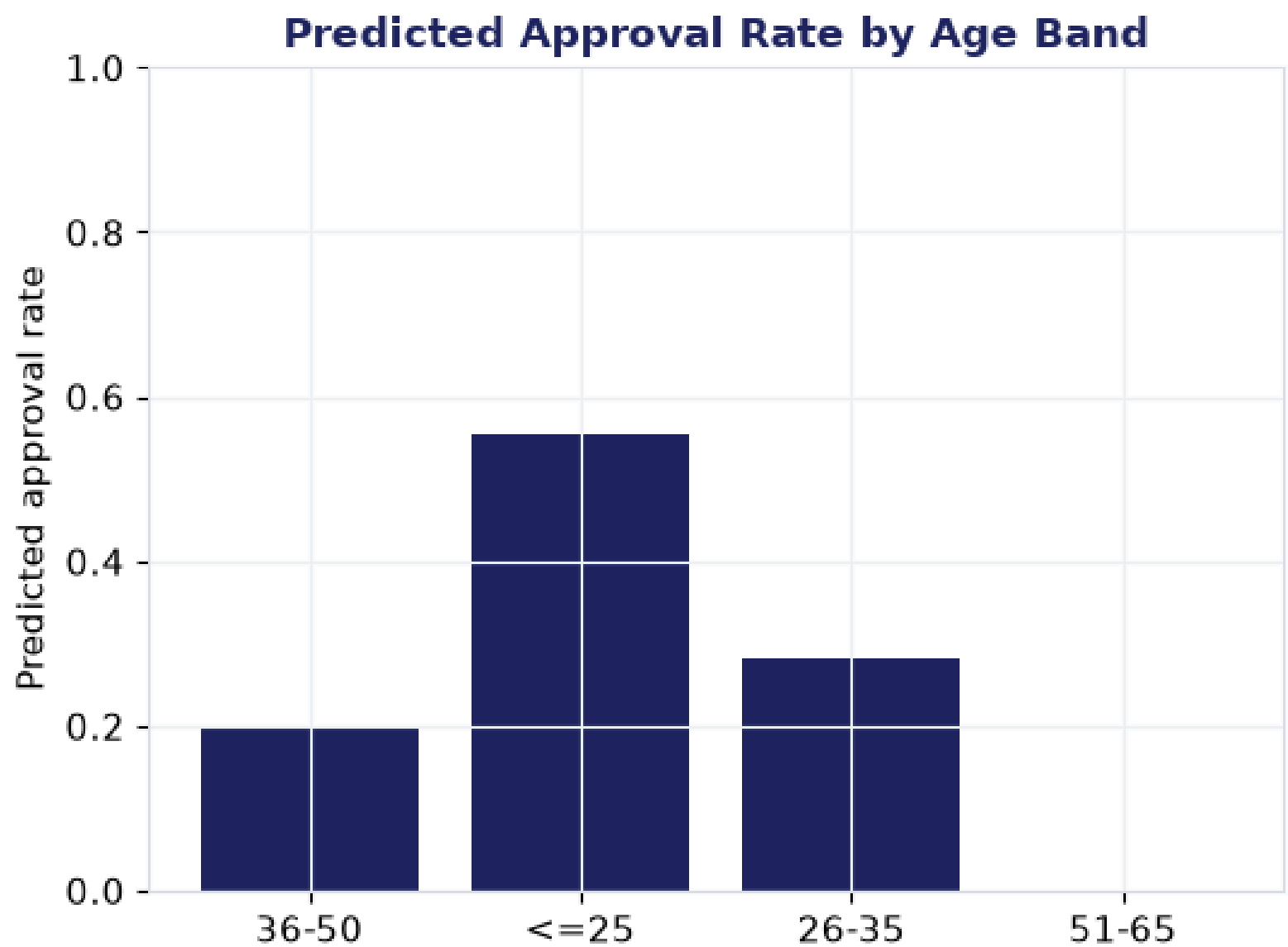
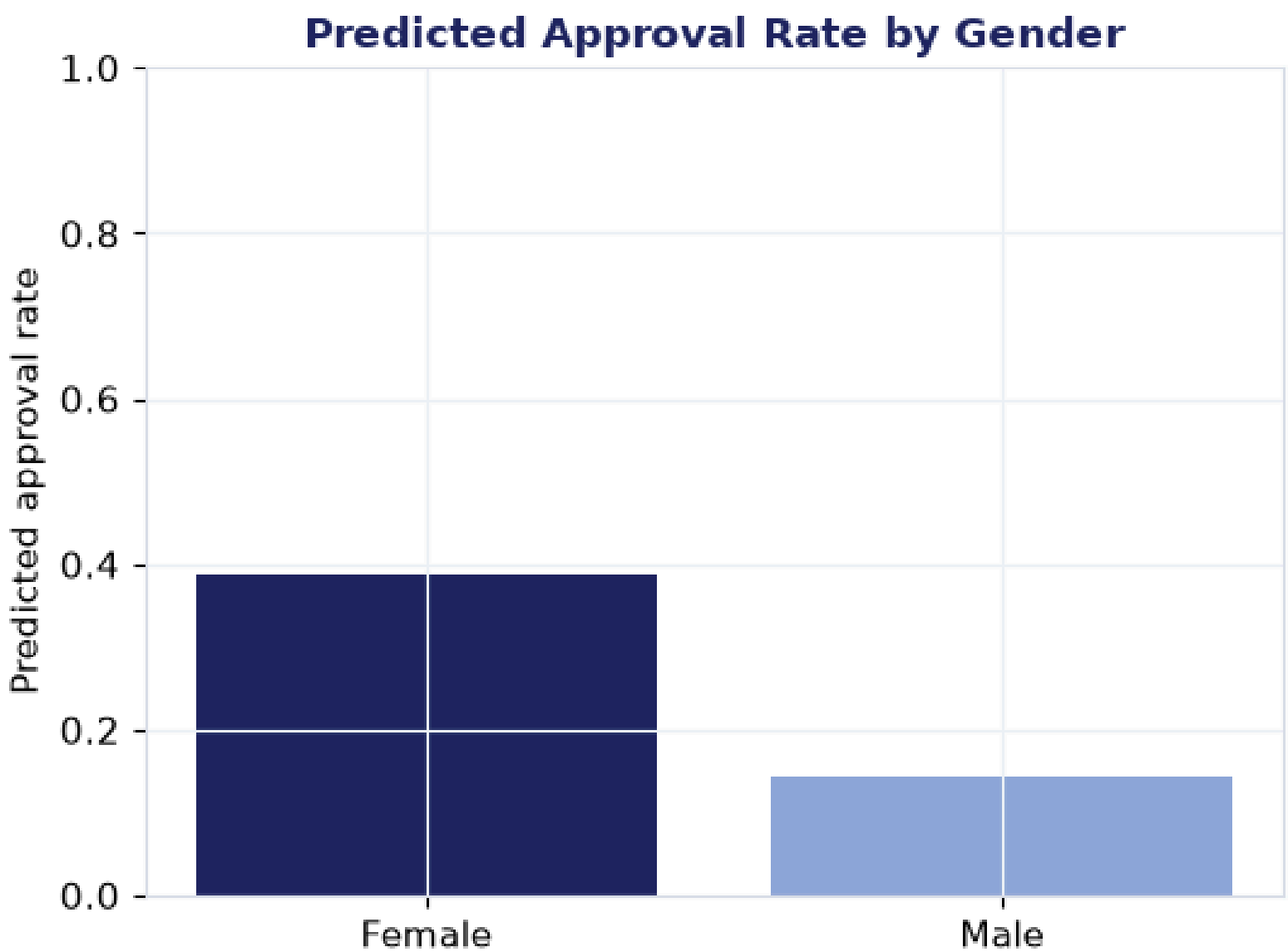
- Our model have identified 1152 customers who we can approach with a loan and have high chance of accepting.

Model Score vs. Business-Rule Quadrant (Never-Borrowed Prospects)



ID	customer_segment	KPI_action_quadrant	model_propensity_score
TAZDLD5N	Deposit-Only Prospect	Segment_C_Untapped_Potential	0.9988595287118065
WKYSXYK6	Deposit-Only Prospect	Segment_C_Untapped_Potential	0.9988531321493297
FI58QUTS	Deposit-Only Prospect	Segment_D_Seasonal_Borrower	0.9984268791734406
PG5PQCC5	Deposit-Only Prospect	Segment_C_Untapped_Potential	0.9976438394120002
SHLWYLW0	Deposit-Only Prospect	Segment_C_Untapped_Potential	0.9968083524726566
LAZ2IB6N	Deposit-Only Prospect	Segment_C_Untapped_Potential	0.9963481206535448
EDFT9KMR	Deposit-Only Prospect	Segment_D_Seasonal_Borrower	0.9962939417348405
RJR0L9NE	Deposit-Only Prospect	Segment_D_Seasonal_Borrower	0.9958225710121771
OM0XSF5P	Deposit-Only Prospect	Segment_D_Seasonal_Borrower	0.9951211831606902
QXJDJKTT	Deposit-Only Prospect	Segment_C_Untapped_Potential	0.9949794740548324

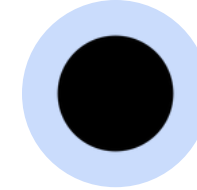
Is there Bias in the Data?



Fairness Report



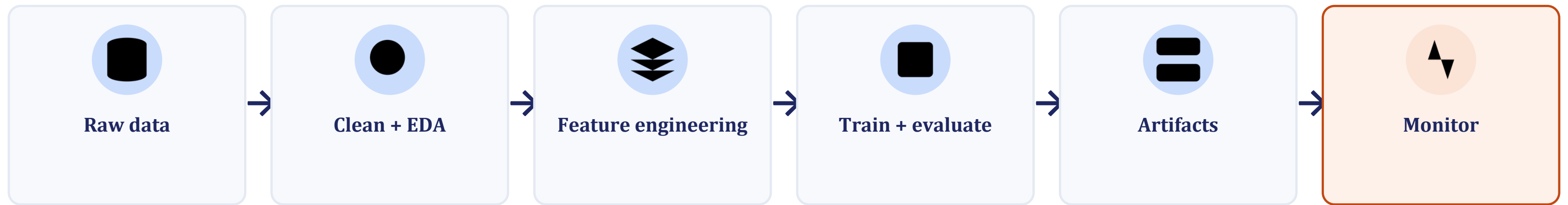
1. Disparate impact



2. Does gender earn its place?

- `disparate_impact_ratio_gender`: 0.35
- 'Gender' adds little to no measurable predictive power here;
 - excluding it from the live model costs little and removes a protected characteristic.

How this runs end to end



GitHub : <https://github.com/Chuene100/Loan-Propensity-Model>

What we watch once this is live



Prediction volume & latency

Every /score call is counted and timed — catches a serving outage or a slow-down before a downstream team notices.



Score distribution

A histogram of propensity scores over time — a sudden shift (everyone scoring high, or low) is often the first sign something upstream changed.



Feature drift (PSI)

Population Stability Index compares today's feature distributions to training — <0.10 stable, 0.10–0.25 monitor, >0.25 consider retraining.



Fairness over time

The disparate-impact check from slide 10 isn't a one-off — re-run on a schedule so a shift in approval-rate gaps gets caught, not just checked once at launch.

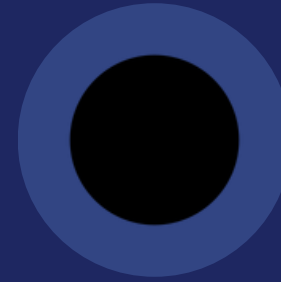


Prometheus + Grafana

The serving app exposes a /metrics endpoint. Grafana connects via a provisioned Prometheus data source, with a starter dashboard covering request volume, latency, error rate, and score distribution.

Recommendations & next steps

- 1** Run on the full historical dataset
- 2** Pilot on a small, monitored segment
- 3** Decide the serving path: batch scoring vs real-time endpoint
- 4** Schedule fairness & drift re-checks



Discussion

Questions!!!